Project Design Phase-II

Data Flow Diagram & User Stories

Date	15 October 2026
Team ID	SkillWallet — ServiceNow Project
Project Name	Automated Employee Onboarding & Offboarding System
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) is a visual representation of information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 — Context Diagram

HR (Requestor)	**Submit Request Status Updates**	**ServiceNow Onboarding System Automated Employee Onboarding & Offboarding**
Manager (Approver)	**Approve / Reject Approval Notification**	**IT, Facilities, Security Task Assignment / Task Completion**

External entities: HR (Requestor), Manager (Approver), and IT, Facilities, Security (Fulfillers).

DFD Level 1 — Detailed Data Flow

The workflow starts when HR submits the Service Catalog form. ServiceNow creates request records, retrieves catalog variables, creates an Employee Lifecycle record, and requests manager approval. Approved requests generate three fulfilment tasks; rejected requests are logged and communicated to HR.

Process / Step	Input / Condition	Output / Data Store
1.0 Service Catalog Intake	HR submits form	D1: sc_request (REQ) D2: sc_req_item (RITM)
2.0 Get Catalog Variables (Flow Action #1)	Requested item record; employee_id, department, manager, joining_date	Variables passed to lifecycle record creation
3.0 Create Lifecycle Record (Flow Action #2)	Employee details	D3: u_employee_lifecycle Status = New
4.0 Ask For Approval (Flow Action #3)	Approval request	D5: sysapproval_approver; Manager approves or rejects
5.0 Create 3 Tasks (Flow Actions #5, #6, #7)	Approved request	D4: sc_task — IT, Facilities, Security tasks
6.0 Wait For Condition (Flow Action #8)	Fulfillers complete tasks	Wait until all tasks are Closed Complete
7.0 Update Lifecycle (Flow Action #9)	All tasks complete	D3 status = Completed
8.0 Close RITM (Flow Action #10)	Lifecycle completed	Requested item closed
9.0 Send Email (Flow Action #11)	Completion	D6: sys_email; notify HR
10.0 Log Rejection (Flow Actions #13–15)	Manager rejects request	Lifecycle status = Rejected; RITM Closed Incomplete; rejection email sent to HR

Data Stores

ID	Data Store	Purpose
D1	sc_request	Stores the top-level request (REQ)
D2	sc_req_item	Stores the requested item (RITM)
D3	u_employee_lifecycle	Custom table storing onboarding/offboarding data
D4	sc_task	Catalog tasks assigned to IT/Facilities/Security
D5	sysapproval_approver	Approval records for manager approval
D6	sys_email	Email notifications sent

Data Flow Summary

#	From	To	Data
1	HR	Service Catalog	Form submission (13 variables)
2	Catalog	sc_request	REQ created
3	Catalog	sc_req_item	RITM created
4	RITM	Flow Designer	Trigger fires
5	Flow	u_employee_lifecycle	Lifecycle record created
6	Flow	sysapproval_approver	Approval requested from manager
7	Manager	sysapproval_approver	Approval / Rejection
8	Flow	sc_task	3 tasks created
9	Fulfillers	sc_task	Tasks closed
10	Flow	u_employee_lifecycle	Status updated to Completed
11	Flow	sc_req_item	RITM closed
12	Flow	sys_email	Notification sent

User Stories

The following user stories define the expected behaviour and acceptance criteria for the Automated Employee Onboarding & Offboarding System.

US-01: HR Submits Onboarding Request

As an HR Manager, I want to submit an onboarding request through a single Service Portal form with all required fields (Employee ID, Department, Manager, Joining Date), so that all relevant departments are notified automatically without me sending emails.

Acceptance Criteria: Single form with 13 fields • Mandatory fields enforced • Submit button generates REQ • Request Summary shows REQ number

US-02: Manager Approves Request

As a Manager, I want to receive an approval notification with all request details, so that I can approve or reject onboarding requests for my team quickly.

Acceptance Criteria: Email notification received • Approve/Reject available in ServiceNow • Decision recorded with timestamp

US-03: IT Receives Auto-Assigned Task

As an IT Administrator, I want to automatically receive a task titled "Create accounts & assign hardware" when a request is approved, so that I can provision accounts and hardware without manual coordination.

Acceptance Criteria: Task auto-created on approval • Assigned to IT – Account Provisioning group • Short description clearly states action

US-04: Facilities Receives Workspace Task

As a Facilities Staff member, I want to receive a task titled "Prepare workspace" with the joining date, so that I can prepare the workspace on time.

Acceptance Criteria: Task auto-created on approval • Assigned to Facilities – Workspace group • Joining date visible on task

US-05: Security Receives Access Task

As a Security Officer, I want to receive a task titled "Issue ID badge & building access", so that I can issue the badge before the employee's first day.

Acceptance Criteria: Task auto-created on approval • Assigned to Security – Access group • Short description clearly states action

US-06: Lifecycle Status Updates Automatically

As an HR Manager, I want the Employee Lifecycle record to update its Status automatically, so that I can see the real-time progress of each request without asking anyone.

Acceptance Criteria: Status = New on submission • Status = Completed when all 3 tasks close • No manual updates needed

US-07: HR Receives Completion Notification

As an HR Manager, I want to receive an email when onboarding completes, so that I know the new hire is ready for day 1.

Acceptance Criteria: Email sent automatically on completion • Subject: "Onboarding complete" • Contains request details

US-08: Rejected Requests Handled Cleanly

As an HR Manager, I want rejected requests to close with a clear status and notification, so that I know the request was not approved and can act accordingly.

Acceptance Criteria: Lifecycle Status = Rejected • RITM = Closed Incomplete • Rejection email sent

US-09: Admin Controls Data Access

As a ServiceNow Administrator, I want role-based ACLs on the Employee Lifecycle table, so that only authorized users can view or modify sensitive data.

Acceptance Criteria: ACLs on read/write/create/delete • Requires admin role • Data protected from unauthorized access

US-10: SLA Tracking on Tasks

As a Service Manager, I want SLAs attached to each catalog task, so that response times per department can be measured and improved.

Acceptance Criteria: SLA on IT task = 1 day • SLA attached automatically • Breach notifications enabled

US-11: Dynamic Form Based on Request Type

As an HR Manager, I want the form to show only relevant fields based on Request Type (Onboarding vs Offboarding), so that the form is clean and free of irrelevant fields.

Acceptance Criteria: Onboarding → Joining Date shown • Offboarding → Exit Date shown • Uses Catalog UI Policies

US-12: Audit Trail for Compliance

As a Compliance Officer, I want every action (approval, task creation, completion) to be logged, so that I can audit the process at any time.

Acceptance Criteria: Flow Execution Details available • Approval history recorded • Task state changes logged • Emails logged

User Story Summary Table

ID	User Story	Priority
US-01	HR submits onboarding request via portal	High
US-02	Manager approves request	High
US-03	IT receives auto-assigned task	High
US-04	Facilities receives workspace task	High
US-05	Security receives access task	High
US-06	Lifecycle status updates automatically	High
US-07	HR receives completion notification	Medium
US-08	Rejected requests handled cleanly	High
US-09	Admin controls data access via ACLs	High
US-10	SLA tracking on tasks	Medium
US-11	Dynamic form based on Request Type	Medium
US-12	Audit trail for compliance	High

Outcome: The DFDs show how requests, approvals, fulfilment tasks, lifecycle records, and notifications move through ServiceNow. The 12 user stories describe the required user-facing behaviours, controls, service levels, and audit capabilities.